

Kothe Viswanath

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San Francisco, CA

Education

Syracuse University

Master of Science in Computer Engineering

Syracuse, NY

Aug 2023 – May 2025

Experience

AI Engineer

Syracuse University

Jun 2025 – Present

- Built **complex multi-agent workflows** in **Python** with **LangGraph**, **MCP** enabling automated, multi-step workflows and automated task execution for automated fuzzing of unknown devices.
- Built **large-scale data pipelines** (**Airflow**, **Kafka**, **Python**) processing 5M+ code files into a **RAG knowledge base** with lineage tracking on **Qdrant DB**.
- Deployed multiple **async FastAPI/FastMCP servers** as **HTTP deployed** in **AWS VPC** and **stdio endpoints**, supporting scalable, distributed workflow execution.
- Designed **DeepEval** pipelines **evaluating** BLEU, ROUGE, recall@k improving **retrieval accuracy** by **~28%**.
- **Collaborated cross-functionally** to define requirements, implement **guardrails**, and design automated **evaluation suites**, improving **reliability and safety**.

Software Engineer

Indian Space Research Organisation (ISRO), Bangalore, India

Jan 2022 – Jun 2023

- Created an **AI/ML powered anomaly detection backend** (LSTM/XGBoost) with **Python**, **FastAPI**, that lowered manual monitoring effort by **40%**, with **MongoDB** used to maintain audit logs and traceability.
- Developed **concurrent Golang Kafka pipelines**, for high throughput data acquisition from multiple devices and services enabling global source of truth for Flink and Spark consumers.
- Built a **FullStack (React, python)** monitoring system **aggregating data** from **REST APIs**, **DBs**, and **WebSockets**, powering anomaly detection and real-time alerts, achieving **sub 50 ms latency**.
- Built a remote-control module with **RBAC**, enabling automated management of antenna systems with **gRPC**.
- Led **cross-team collaboration** for CI/CD pipelines, implementing **Git/ArgoCD based IaC on Kubernetes**,
- Automated **testing**, **ML retraining**, ensuring reliable and fast deployments ensuring **KPIs** like reliability.
- Implemented observability with **OpenTelemetry**, **Prometheus** and **Grafana**, improving **incident RCA by 60%**.

Software Engineer

Nuviso Networks, Bangalore, India

Jun 2020 – Dec 2021

- Improved **network data collection module** enabling global monitoring of switches, routers, and edge devices, reducing packet processing time and latency by almost 15% (*Python, Go, SNMP, Netmiko, Nmap*).
- Implemented **network monitoring and logging frameworks** with automated testing, improving uptime, traceability, and fault detection (*Prometheus, Grafana, Pytest, custom monitoring scripts*).
- Conducted **network security audits and vulnerability scanning**, identifying and resolving configuration and protocol issues to maintain secure operations (*Nmap, Wireshark, Python scripting, SIEM*)..

Technical Skills

Languages & frameworks: Go, Python, SQL, Typescript, FastAPI, Go mux, Spring Boot, REST, gRPC, WebSockets

Tools: Kafka, Flink, Spark, MongoDB, postgresql, Qdrant, RBAC, Grafana, ELK, OpenTelemetry, Linux, TCP/IP

Testing & CI/CD: PyTest, JUnit, Go test, GitHub Actions, Jenkins, Docker, Kubernetes, ArgoCD, AWS, EKS

LLM & Agentic AI: GPT, Claude, MCP, LangSmith, LangGraph, fastMCP, RAG, RAGAS, guardrails

Personal Projects

JarVish: <https://github.com/vishwa419/JarVish>

Oct 2025 – Present

Jarvis clone, powered by **GPT**, **deepagents**, **subagents**, **RAG** and **MCPs** as **microservice**. Powering end to end daily workflow automation with observability (LangChain, LangGraph, Qdrant, Docker, Opik).

AGENTIC DOCKER: https://github.com/vishwa419/Kitty_containers

Aug 2025 – Sep 2025

Built a modular **agentic-AI system** for task orchestration, integrating GPT, MCP and RAG to automate container deployments (Python, LangGraph, LangChain, Qdrant).

MLOPS: https://github.com/vishwa419/MLOps_Inference

Jun 2025 – Aug 2025

GitOps-driven ML pipeline for automated data validation, feature store, model training, deployment, and monitoring across environments (Python, BentoML, MLFlow, Docker, Grafana, Great Expectations, Feast).

Novel Algorithm for explainable CNNs: <https://doi.org/10.1109/ICCCNT56998.2023.10306989>